Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 10)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 1 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 2 . 1 Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

Waste category	Example / source	Storage or disposal method	
High-level waste			
Intermediate-level waste			
Low-level solid waste			

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0 2 . 2 Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

0 2 . 3 Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

10

0 3 Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO_2 over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NOx)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NOx and hydrocarbons	Days-weeks	-

0 3 . 1 Explain the natural greenhouse effect.

[3 marks]

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0 3 . 2 Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

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0 3 . 3 Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO_2 emissions by the same mass.

[2 marks]

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0 3 . 4 Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

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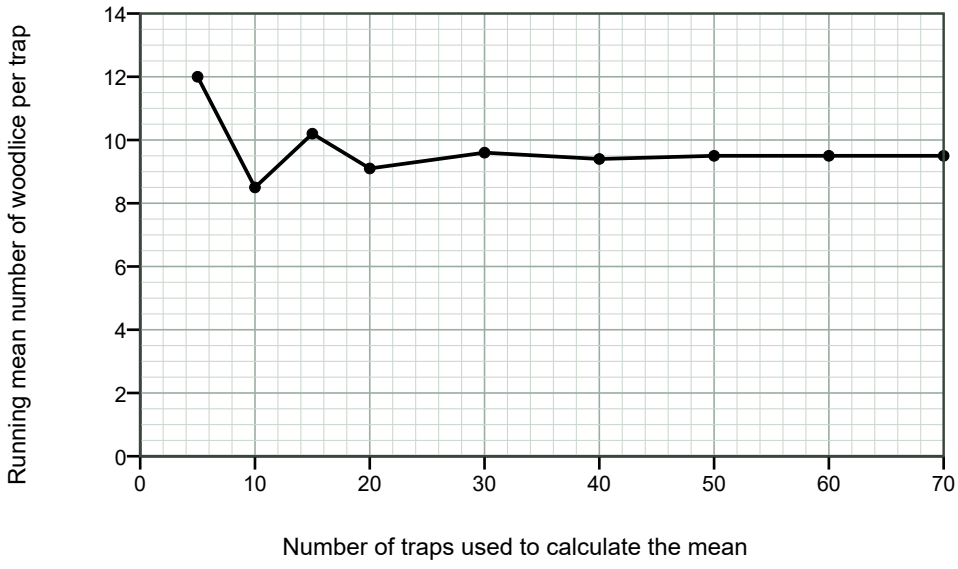
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0 4 Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 4 . 1 Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

0 4 . 2 Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

0 4 . 3 The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

0 4 . 4 Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

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0 5 A pumped-storage HEP station has an upper reservoir 500 m above its turbines. It can hold 7 million m^3 of water. The density of water is 1000 kg m^{-3} and g = 9.8 N kg^{-1}.

0 5 . 1 Describe how pumped-storage HEP is used for peak shaving.

[3 marks]

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0 5 . 2 Calculate the maximum gravitational potential energy stored when the upper reservoir is full. Give your answer in standard form.

[3 marks]

Answer ..... J

0 5 . 3 The station returns 75% of the electricity used for pumping. Explain why the process is still worthwhile despite this loss.

[2 marks]

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0 5 . 4 Give **two** locational requirements of a pumped-storage station.

[2 marks]

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0 6 A dam and reservoir are to be built on a river to supply water and regulate flow.

0 6 . 1 Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

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0 6 . 2 Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

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0 6 . 3 Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

[2 marks]

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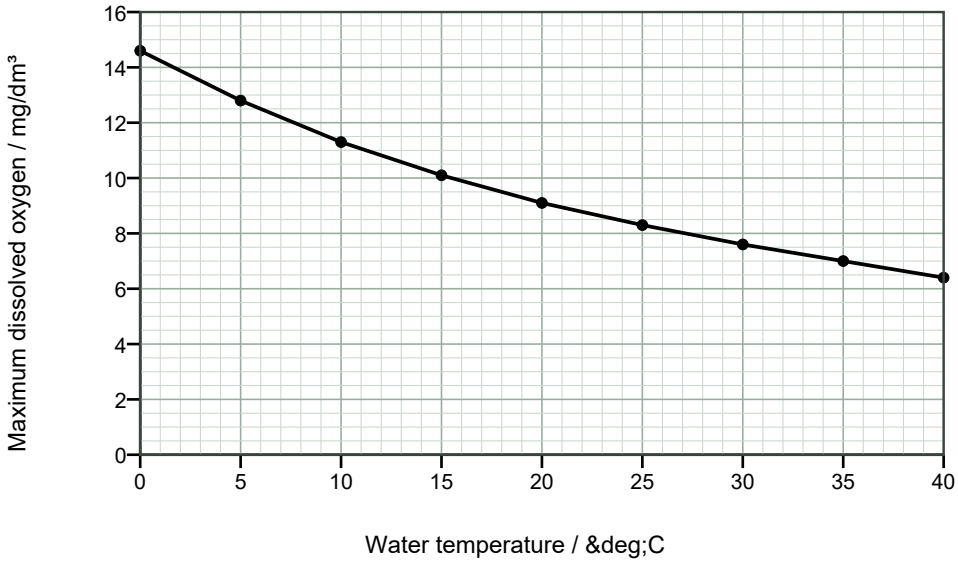
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0 7 Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3}.

Figure 1



0 7 . 1 Describe the relationship shown in Figure 1.

[1 mark]

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0 7 . 2 Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Answer ____ °C

0 7 . 3 In summer the river upstream of the power station is at 20 °C and the effluent raises the temperature by 7 °C. Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

Answer ____ mg/dm³

0 7 . 4 Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

[2 marks]

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0 7 . 5 Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

[3 marks]

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0 7 . 6 Describe how a cooling tower reduces the temperature of power station effluent water.

[1 mark]

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0 8 . 1 Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

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0 8 . 2 Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

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0 8 . 3 Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

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0 9 . 1 Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

0 9 . 2 Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

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0 9 . 3 Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

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1 0 . 1 Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS